



Master Presenter's Manual

for

R Programming

Curriculum Code: OV-1521

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1. Introduction

This Presenter's Manual is to be used for the **R Programming** course.

After the completion of this course, students will be able to:

- ❖ Master data exploration, data visualization, predictive analytics and descriptive analytics techniques with the R language.

2. General Information on Session Schedule

The table below shows the details of the modules, the number of hours being spent on the module along with a break-up of the hours being spent on the module.

This course is based on theory and practice lab session. A session is of two hours duration.

Module Wise Session Allocation

Module	Online Hours	Skill Code
Programming in R	40	1262

Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

The table below shows the details of the module, the number of hours being spent on the module along with a break-up of the hours being spent on each session.

3. Week Wise Schedule

Topic \ Week		1	2	3	4	5
Programming in R	D1	TL1	TL5	TL9	TL13	TL17
	D2	TL2	TL6	TL10	TL14	TL18
	D3	TL3	TL7	TL11	TL15	TL19
	D4	TL4	TL8	TL12	TL16	TL20

Where,

D = Day, TL= Online Session

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	R – TL1	All the topics as listed below from Session 1 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 1: Introduction to Data Science</u> - Describe the concept of Data Science - Describe the life cycle of Data Science - List the tools of Data Science - Describe business analytics - Describe types of business analytics - Identify the application of business analytics	Programming Guide for R SG-Session 1 XP-Session 1 TG-Session 1
2	R – TL2	All the topics as listed below from Session 2 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 2: Introduction to R Programming</u> - Describe Object Oriented Programming (OOP) - Define basic OOP concepts - Describe functional programming in R	Programming Guide for R SG-Session 2 XP-Session 2 TG-Session 2
3	R – TL3	All the topics as listed below from Session 3 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 3: Basics of R Programming</u> - Describe the importance of R - Describe R features - Explain data types and variables in R - Explain operators in R - Describe string functions in R - Describe conditional statements in R - Explain Loops and functions in R	Programming Guide for R SG-Session 3 XP-Session 3 TG-Session 3
4	R – TL4	All the topics as listed below from Session 4 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 4: R Data Structures and Apply Series</u> - Describe data structures - Identify data structures - Assign and apply values to data structure - Manipulate data - Describe R Apply function series	Programming Guide for R SG-Session 4 XP-Session 4 TG-Session 4

Session No.	Session Title	Session Details	Deliverables' Mapping
5	R – TL5	All the exercises/codes of Session 1 to Session 4 of <i>Programming Guide for R</i> book should be covered in this session.	Programming Guide for R Session 1-4
6	R – TL6	All the topics as listed below from Session 5 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 5: Data Visualization I</u> - Describe data visualization - Describe data visualization using graphics in R	Programming Guide for R SG-Session 5 XP-Session 5 TG-Session 5
7	R – TL7	All the topics as listed below from Session 6 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 6: Data Visualization II</u> - Describe ggplot2 - Describe R Shiny - Describe different file formats of graphic outputs	Programming Guide for R SG-Session 6 XP-Session 6 TG-Session 6
8	R – TL8	All the topics as listed below from Session 7 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 7: Statistics for Data Science I</u> - Describe probability - Describe importance of statistics for data science - Describe random variables - Explain discrete and continuous probability distributions	Programming Guide for R SG-Session 7 XP-Session 7 TG-Session 7
9	R – TL9	All the topics as listed below from Session 8 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 8: Statistics for Data Science II</u> - Describe data sampling - Define parameter and statistic - Describe hypothesis and its types - Describe confidence interval and level of significance - Identify the various hypothesis tests	Programming Guide for R SG-Session 8 XP-Session 8 TG-Session 8
10	R – TL10	All the exercises/codes of Session 5 to Session 8 of <i>Programming Guide for R</i> book should be covered in this session.	Programming Guide for R Session 5-8

Session No.	Session Title	Session Details	Deliverables' Mapping
11	R – TL11	All the topics as listed below from Session 9 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 9: Linear Regression Analysis</u> - Describe machine learning - Describe regression analysis and its types - Describe linear regression	Programming Guide for R SG-Session 9 XP-Session 9 TG-Session 9
12	R – TL12	All the topics as listed below from Session 10 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 10: Non-Linear Regression Analysis</u> - Describe non-linear regression - Describe the important types of non-linear regression	Programming Guide for R SG-Session 10 XP-Session 10 TG-Session 10
13	R – TL13	All the exercises/codes of Session 9 and Session 10 of <i>Programming Guide for R</i> book should be covered in this session.	Programming Guide for R Session 9 & 10
14	R – TL14	All the topics as listed below from Session 11 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 11: Classification I</u> - Describe classification - Describe Support Vector machines - Describe K- nearest neighbours	Programming Guide for R SG-Session 11 XP-Session 11 TG-Session 11
15 & 16	R – TL15 & TL16	All the topics as listed below from Session 12 and Session 13 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 12: Classification II</u> - Describe Naïve Bayes classifier - Describe Decision Tree classifier - Describe Random Forest classifier <u>Session 13: Evaluating Classifier Models</u> - Describe Naïve Bayes classifier - Describe Decision Tree classifier - Describe Random Forest classifier	Programming Guide for R SG-Session 12 & 13 XP-Session 12 & 13 TG-Session 12 & 13

Session No.	Session Title	Session Details	Deliverables' Mapping
17	R – TL17	All the exercises/codes of Session 11 and Session 12 of <i>Programming Guide for R</i> book should be covered in this session.	Programming Guide for R Session 11-13
18 & 19	R – TL18 & TL19	All the topics as listed below from Session 14 and Session 15 of <i>Programming Guide for R</i> book should be covered in this session. <u>Session 14: Clustering</u> - Describe clustering - Describe various methods of clustering - Understand k-means clustering - Describe hierarchical clustering <u>Session 15: Association</u> - Describe association - Describe association rule - Describe Apriori algorithm	Programming Guide for R SG-Session 14 & 15 XP-Session 14 & 15 TG-Session 14 & 15
20	R – TL20	All the exercises/codes of Session 13 to Session 15 of <i>Programming Guide for R</i> book should be covered in this session.	Programming Guide for R Session 13-15

5. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

➤ Deliverables for Students:

Learner's Guide

eBook consisting of the knowledge to be imparted (wherever applicable)

Assignments

Case study/scenario based problems to be solved by learners

Article

Additional knowledge in the form of articles that are referred from Web

Practice Questions

Additional practice questions for learners at the end of each session

Module	Book Code	Book Name
KIT	OV-1521-KIT01-01-A	Kit of 1 title for OV-1521 R Programming
Data Science using R Programming	AINNPGUDR10723E000	Programming Guide for R

➤ **Deliverables for Faculty**

Presenter's Manual	A training delivery plan for the trainers
Trainer's Guide	A session wise guide for the trainers
Classroom Presentations	Session wise PowerPoint presentations as teaching aid, wherever applicable

Module Name	Book Code	Book Name
KIT	OV-1521-KIT01-01-A	Kit of 1 title for OV-1521 R Programming
Data Science using R Programming	AINNPGUDR10723E000	Programming Guide for R

6. Library References

➤ Beginning R: The Statistical Programming Language by Mark Gardener
➤ R Programming for Beginners by Sandip Rakshit
➤ R for data science : Import, Tidy, Transform, Visualize, And Model Data by Hadley Wickham, Garrett Golemund

7. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Final Examination (Objective)
- Practical Examination

7.1 Award of Grades

Overall Weighted Marks $\geq 40\%$ but $< 60\%$ qualifies for PASS

Overall Weighted Marks $\geq 60\%$ but $< 75\%$ qualifies for CREDIT and

Overall Weighted Marks $\geq 75\%$ qualifies for DISTINCTION

Note: To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in each of the elements of the Examination; otherwise he/she will be declared as 'Referred'.

7.2 Final Examination 1 (E1)

Below is the list of modules to be covered in Final Examination along with their weightages:

Final Examination 1 (E1)

Marks: 50

Duration: 1 hour

Type: Objective

S#	Name of the Subject	Marks
1.	Data Science using R Programming	50

Final Examination 2 (R1)

Marks: 50

Duration: 2 hours

Type: Practical

S#	Name of the Subject	Marks
1.	Data Science using R Programming	50

Evaluation Details

OES-50%

Practical-50%

8. Classroom and Lab Pre-requisites

The requirements for conducting the sessions have been described as follows:

8.1 Software Requirements

Microsoft Windows 11
R Studio latest version

8.2 Recommended Hardware

Intel Core i5/i7 Processor or higher
8 GB or 16 GB RAM
1 TB Hard Disk space
Mouse
Keyboard

8.3 Other Requirements (For Classroom)

Computer for giving demonstrations wherever applicable
LCD projector

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